A Semisupervised Siamese Network for Efficient Change Detection in Heterogeneous Remote Sensing Images

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***Abstract*— Change detection in heterogeneous remote sensing images is crucial for emergencies, such as disaster assessment. Existing methods based on homogeneous transformation suffer from the high computational cost that makes the change detection tasks time-consuming. To solve this problem, this article presents a new semisupervised Siamese network (S^3N) based on transfer learning. In the proposed S^3N, the low- and deep-level features are separated and treated differently for transfer learning. By incorporating two identical subnetworks that are both pretrained on natural images, the proposed S^3N eliminates the computational cost for learning the low-level features that are universal for both remote sensing images and natural images. As the deep-level features contain different semantics between remote sensing images and natural images, a novel transfer learning strategy is presented to train only the weights of the layers for deep-level features in the proposed S^3N. The decrease in the number of network parameters to be trained reduces the demand for training samples, leading to a significant decrease in computational cost. Afterward, the thresholding method, *Otsu*, is applied to the difference map derived by the proposed S^3N to obtain the final binary map of change detection. Three data sets including different types of heterogeneous remote sensing images are employed to evaluate the performance of the proposed S^3N. The experimental results demonstrate that the proposed S^3N can achieve a comparable detection performance with much lower computational cost, compared with state-of-the-art change detection algorithms.**

***Index Terms*— Change detection, heterogeneous, remote sensing, transfer learning.**

I. Introduction

CHANGE detection in remote sensing images is to monitor the changes on the surface of Earth that are crucial for emergencies, such as disaster assessment [1], [2]. Compared with homogeneous remote sensing images that are difficult to collect promptly, heterogeneous remote sensing images are more accessible and used commonly for timely change detection in emergency cases.

For change detection in heterogeneous remote sensing images, direct comparison operations (e.g., difference) commonly used for homogeneous images [3]–[11] are not applicable. To address this issue, various methods are proposed [12]–[25], [28], [29]. Recent studies [18]–[25], [28], [29] show that homogeneous transformation is capable of achieving change detection in heterogeneous remote sensing images. In homogeneous transformation, heterogeneous remote sensing images are transformed into an identical feature space that can be compared directly. Compared with traditional methods, such as post-classification comparison [12]–[14] and stacked images joint-detection [15], [16], the methods-based homogeneous transformation are more intuitive and universal [17].

Earlier methods [18], [19] based on homogeneous transformation use imaging parameters of remote sensing satellites to predict the presence of preevent ground objects in the scene of post-event heterogeneous images. The predicted presence is compared directly with that in the same position of post-event images to derive the change detection results. As the imaging parameters of satellites are normally inaccessible, later research [20]–[25], [28], [29] prefers machine learning for homogeneous transformation. Liu *et al.* [20], [21] proposed a multivalue estimation method based on K-nearest neighbors techniques to map pixels between heterogeneous images for homogeneous transformation. Gong *et al.* [22] established mapping relationship between multisource data based on coupled dictionary learning. It is later improved by incorporating a Siamese network with a symmetric structure that facilitates the extraction of homogeneous features from heterogeneous remote sensing images by the same layers of the two identical subnetworks [23]. In our previous research [24], [25], an iterative strategy is proposed to refine the transformed images based on image style transfer (IST) with Siamese structure [27] for change detection in heterogeneous optical and synthetic aperture radar (SAR) images. Liu *et al.* [28] employed the classification on heterogeneous optical and SAR data based on a multimapping value strategy and a belief fusion theory to manage uncertain information. Other machine learning-based algorithms, such as Bayesian nonparametric method [29] and

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kernel canonical correlation [30], are exploited based on homogeneous transformation for change detection.

The above methods achieve remarkable detection results in heterogeneous remote sensing images. However, most of them suffer from high computational cost in the online training procedure. For change detection in heterogeneous remote sensing images [20]–[25], [28], [29], only online training is feasible because of lack of massive training samples. In other words, for each case of heterogeneous remote sensing images, the online training procedure is essential. Accordingly, the training procedures in the existing methods with high computational cost decrease the efficiency of change detection significantly, which are critical for emergencies.

In the online training procedures, the computational complexity depends on the sizes of training samples and the number of iterative steps. The existing methods learn all the homogenous features from scratch for the transformation of heterogeneous remote sensing images and thus require massive training samples and iterative steps. In [23]–[25], as the Siamese networks used for homogeneous feature extraction include two subnetworks with massive network parameters to be trained, the learning of all the homogenous features requires entire images for training. Because remote sensing images are normally with large sizes, the online training procedures have extremely high computational cost. In [20] and [21], to build multivalue mapping for pixel-level homogenous transformation, a sorting operation based on K-nearest neighbors technique is applied to training patches selected from the heterogeneous remote sensing images. The expansion of remote sensing image sizes increases the demand for training patches for the sorting operation and thus the computational cost.

Recently, transfer learning [31] is utilized to improve efficiency for training [32], [33] in case of training sample shortage. However, most of the popular deep neural networks (DNN) [34], [35], [38] for transfer learning are pretrained on natural images not remote sensing images. The direct transfer learning of DNNs pretrained on natural images does not perform well on the change detection in heterogeneous remote sensing images. It is because the deep-level semantic features used for change detection differ significantly between natural images and remote sensing images. As a result, the deep-level features that are directly transferred from natural images cannot accommodate the semantic content of remote sensing images for change detection.

In this article, we present a new semisupervised Siamese network (S^3N) with a novel transfer learning strategy for change detection in heterogeneous remote sensing images. The proposed S^3N incorporates two identical subnetworks pretrained on natural images to separate the low/deep-level features and treats them differently by the novel transfer learning strategy. In the proposed S^3N, the low-level features [24]–[26] that refer to textures, edges, and corners are universal for both natural images and remote sensing images. As a result, the pretrained subnetworks on natural images can be utilized directly to extract the low-level features, leading to the reduction of computational cost for training. As mentioned above, compared with the low-level features, the deep-level features that carry semantics for change detection differ significantly between natural images and remote sensing images. Therefore, we present a novel transfer learning strategy to train only the weights of the layers for deep-level features in the proposed S^3N. As only the weights of the layers for deep-level features are adjusted in the training process, the number of network parameters to be learned in the proposed S^3N is decreased, leading to a significant reduction of demand for training samples. With the decrease of both the training samples and network parameters to be trained, the number of iterative steps for online training and thus the computational cost is reduced by the Siamese network, leading to significant improvements in detection efficiency.

Compared with the existing methods of change detection in heterogeneous remote sensing images [20]–[25], the proposed S^3N forms the Siamese network structure trained by the novel transfer learning strategy, leading to a significant reduction of the computational cost for online training. Compared with symmetric convolutional coupling network (SCCN) [23] engaged in a complicated training process, the proposed S^3N embeds in the Siamese structure two pretrained visual geometry group (VGG) subnetworks that are sufficiently deep to separate the deep- and low-level features. The low-level features are shared by both natural images and remote sensing images while the deep-level features can be directly comparable for change detection. As a result, the proposed S^3N shortens the training process significantly and benefits the transfer learning for heterogeneous remote sensing images.

Compared with the conventional fine-tune strategies that either update all the parameters of the network [40] or only fine-tunes the fully connected layers [41], the proposed transfer learning strategy freezes the weights of the layers for low-level features and only trains the weights of the layers for deep-level features. Therefore, the proposed transfer learning strategy can transfer the knowledge of the low-level features shared by both remote sensing images and natural images while only needs to adjust the deep-level features to the change detection task in heterogeneous remote sensing images. In addition, to the best of our knowledge, the proposed S^3N is the first achievement of transfer learning on change detection in heterogeneous remote sensing images. Compared with the existing methods of change detection in heterogeneous remote sensing images [20]–[25], the novel transfer learning strategy discriminates the low- and deep-level features for homogeneous transformation, greatly reducing the number of parameters to be trained. The time-consuming training procedures in the existing methods are then avoided. As a result, the proposed transfer learning strategy decreases the computational cost significantly and thus greatly improves the detection efficiency. It is worth noting that the computational cost is also robust to the size variation of heterogeneous remote sensing images because of the constant number of network parameters to be trained by the novel transfer learning strategy.

Afterward, a simple thresholding method, *Otsu* [36], is applied to the difference map derived by the proposed S^3N to obtain the final binary map of change detection. We apply such a simple thresholding method to prove that the improvements

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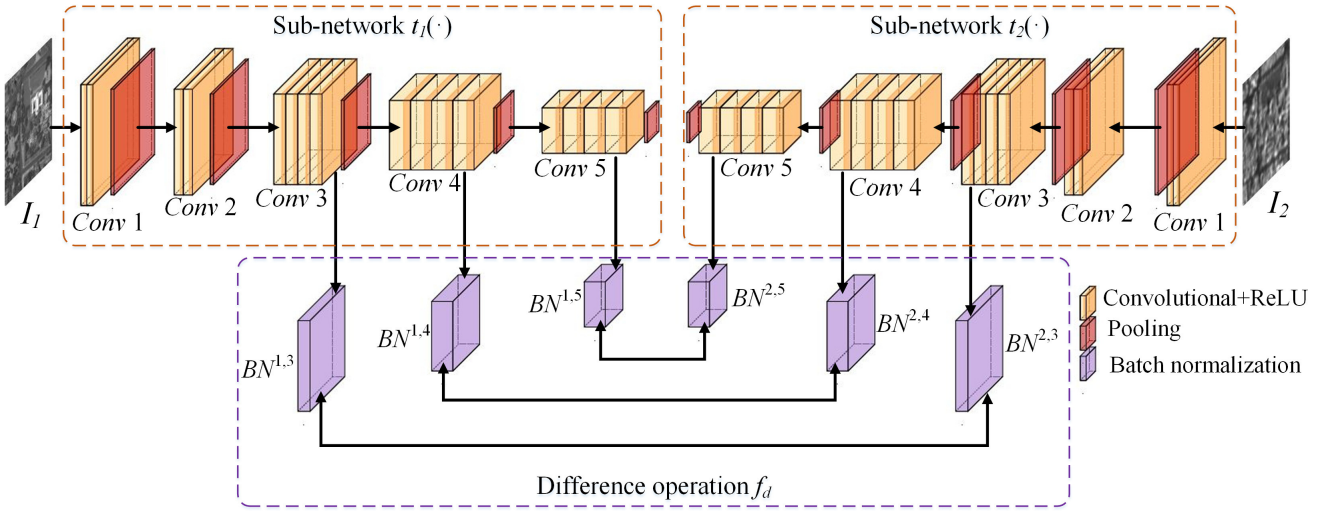


Fig. 1. Structure of the proposed S^{3}N. In the VGG subnetworks, *Conv* stands for the convolutional layers with rectified linear unit (ReLU) operation and *BN* represents batch normalization.

of detection efficiency are mainly generated from the proposed S^{3}N with the novel transfer learning strategy and to neutralize the influence of other factors. Three data sets of heterogeneous remote sensing images, including panchromatic, multispectral, SAR, polarimetric SAR (PolSAR), and Normalized Difference Vegetation Index (NDVI) images, are adopted to validate the proposed S^{3}N. The experiments demonstrate that the proposed S^{3}N is much more computationally efficient and achieves comparable detection performance, compared with state-of-the-art (SOTA) methods.

The remainder of this article is organized as follows. Section II formulates the problem of change detection in heterogeneous remote sensing images and describes the details of the proposed S^{3}N with the novel transfer learning strategy. In Section III, the experimental results are presented. The concluding remarks are given in Section IV.

II. Proposed Semisupervised Siamese Network (S^{3}N)

In this section, we formulate the problem of change detection in heterogeneous remote sensing images. Then, the architecture and formulation of the proposed new S^{3}N that follows the problem formulated in Section II-A are delineated. Next, the novel transfer learning strategy with a new loss function is provided. Finally, we give the inference procedure of deriving the final change detection map, I_b.

A. Problem Formulation

Suppose that I^{pre} and I^{post} are a pair of heterogeneous remote sensing images acquired before (pre-) and after (post-) a change event happens, respectively. Both of I^{pre} and I^{post} are coregistered with each other and rescaled into the same size by bilinear interpolation [37]. We aim to detect the regions of change between I^{pre} and I^{post}. The final detection result is a binary map, named I_b, where the pixels of change and nonchange are labeled as the values of "1" and "0", respectively.

The problem of change detection in heterogeneous remote sensing images is formulated as

$$I_b = f_b(I_d), \tag{1}$$

$$I_d = f_d(t_1(I^{pre}), t_2(I^{post})) \tag{2}$$

where $t_1(\cdot)$ and $t_2(\cdot)$ are the homogeneous transformation functions that map I^{pre} and I^{post}, respectively, into the same feature space. The function $f_d(\cdot,\cdot)$ applies the difference operation that inputs the homogeneous features: $t_1(I^{pre})$ and $t_2(I^{post})$, and outputs the absolute difference to generate the difference map I_d. Finally, the difference map that indicates the change degree is thresholded by f_b to generate the final change detection result, I_b.

B. Network Architecture and Formulation

To perform homogeneous transformation of heterogeneous remote sensing images, the proposed S^{3}N integrates a Siamese structure that maps heterogeneous remote sensing images into homogeneous feature space for direct comparison, as shown in Fig. 1 and Table I.

The proposed S^{3}N includes three parts: two VGG networks with identical structures as the subnetworks and the difference operation f_d. For convenience, we use *Conv*, *Pooling*, and *BN* to represent the convolutional, pooling, and batch normalization layers, respectively. The weights of the layers in the VGG subnetworks have been fully pretrained on the ImageNet [38], including more than ten million colored natural images.

In the proposed S^{3}N, $t_1(\cdot)$ and $t_2(\cdot)$ in (2) are achieved by the two VGG subnetworks. Compared with the AlexNet [34] that is visualized in [42], the VGG subnetwork is much deeper with more convolutional layers and is able to separate the low- and deep-level features, as illustrated in [27]. The two VGG subnetworks take I^{pre} and I^{post} as the inputs, respectively, and map them into homogeneous feature space. Inspired by IST [24], [25], we select the outputs of the convolutional layers of the proposed S^{3}N for deep-level features as the

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TABLE I
PARAMETER DESCRIPTION OF SUBNETWORKS OF THE PROPOSED S^{3}N

Layer Type		#Filters	Size of Filters	#Parameters
Conv 1	*Conv* 1-1	64	$3 \times 3 \times 64$	1.7K
	Conv 1-2	64	$3 \times 3 \times 64$	36K
	Pooling	1	2×2	
Conv 2	*Conv* 2-1	128	$3 \times 3 \times 128$	73K
	Conv 2-2	128	$3 \times 3 \times 128$	147K
	Pooling	1	2×2	
Conv 3	*Conv* 3-1	256	$3 \times 3 \times 256$	300K
	Conv 3-2	256	$3 \times 3 \times 256$	600K
	Conv 3-3	256	$3 \times 3 \times 256$	600K
	Conv 3-4	256	$3 \times 3 \times 256$	600K
	Pooling	1	2×2	
Conv 4	*Conv* 4-1	512	$3 \times 3 \times 512$	1.1M
	Conv 4-2	512	$3 \times 3 \times 512$	2.3M
	Conv 4-3	512	$3 \times 3 \times 512$	2.3M
	Conv 4-4	512	$3 \times 3 \times 512$	2.3M
	Pooling	1	2×2	
Conv 5	*Conv* 5-1	512	$3 \times 3 \times 512$	2.3M
	Conv 5-2	512	$3 \times 3 \times 512$	2.3M
	Conv 5-3	512	$3 \times 3 \times 512$	2.3M
	Conv 5-4	512	$3 \times 3 \times 512$	2.3M
	Pooling	1	2×2	

homogeneous features. To improve efficiency for training the proposed S^{3}N, the novel transfer learning strategy is utilized to train only the weights of the layers for deep-level features. To derive the difference map, the operation f_d in (2) is performed by directly differentiating the layer outputs of homogeneous features between the two VGG subnetworks.

In the change detection of heterogeneous remote sensing images, we aim to detect the change of semantic content. As demonstrated in our previous work [25], the outputs of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, can be used as the deep-level semantic content features. Therefore, they are available for change detection in heterogeneous remote sensing images. Although *Conv* 5-4 is validated as the best layer to describe the semantic content by [25], the pooling layers of VGG networks down-sample its resolutions severely by a factor of 16, resulting in substantial loss of detection details. To mitigate the problem, *Conv* 3-4 and *Conv* 4-4 are added to improve the representations of details. *Conv* 1-2 and *Conv* 2-2 are not considered because they are not abstract enough to represent the deep-level semantic content features [27]. As a result, we select the outputs of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 as effective features for homogeneous transformation.

In the difference operation f_d, we use batch normalization [39] as a routine operation to balance the influence of different scales of semantic content features, i.e., the outputs of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 of the proposed S^{3}N. In the pretrained VGG network [38], there is an enormous difference of variance between the outputs of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4. It makes one of the three layers dominate and the influence of other layers vanish in the difference map. To balance the influence of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 on the difference map, the proposed S^{3}N utilizes batch normalization to refine the three layers of two VGG subnetworks, keeping them in the same ranges for change detection. In the batch normalization [39], the parameters γ and β are defined as the scale and shift parameters, respectively, both of which are used to transform the normalized values for nonlinear activation functions. We freeze the pair of parameters γ and β to be 1 and 0, respectively, the same practice used in [43] and [44]. Therefore, the outputs of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, i.e., the semantic features in different scales, are rescaled into the same ranges with zero mean and unit variance by the Z-Score normalization. As a result, the influence of semantic features in different scales is balanced for homogeneous transformation and thus change detection. The outputs of *BN* layers, i.e., $BN^{1,m}$ and $BN^{2,m}$, $m = 3, 4, 5$, are then used for direct comparison in change detection.

In the proposed S^{3}N, the Siamese structure facilitates the extraction of homogeneous features between the heterogeneous remote sensing images. The two VGG subnetworks with the same structures assure the outputs of symmetric layers of them are comparable with the same sizes and similar properties.

Compared with the existing methods [23]–[25] via Siamese structures, the proposed S^{3}N embeds the two VGG subnetworks capable of collecting the deep-level semantic content features that are directly comparable for change detection. Moreover, with the pretrained VGG subnetworks, the proposed S^{3}N benefits the following transfer learning from natural images to remote sensing images.

C. New Loss Function and Network Transfer Learning Strategy

As the two VGG subnetworks in the proposed S^{3}N are pretrained on natural images, the features are not directly applicable for change detection in heterogeneous remote sensing images. However, as the proposed S^{3}N consists of two VGG subnetworks with massive network parameters, the relearning of all the parameters from scratch is time-consuming.

To solve this problem, the features of pretrained VGG subnetworks in the proposed S^{3}N are separated into two categories: low- and deep-levels for transfer learning. The low-level features, such as textures, edges, and corners, are universal for natural images and remote sensing images, which can be transferred directly from the pretrained VGG networks. Different from the low-level features, the deep-level features are generated by the deep layers of the VGG subnetwork. They refer to the semantic content that differs enormously between natural images and remote sensing images. As a result, the deep-level features cannot be directly transferred and need further learning to accommodate the heterogeneous remote sensing images, I^{pre} and I^{post}.

We present a new loss function with the transfer learning strategy to adjust the deep-level features efficiently for change detection in heterogeneous remote sensing images. The loss function is defined as follows:

$$L_{\mathrm{S}^3\mathrm{N}} = \sum_{m=3}^{5} \left[\frac{\alpha_m}{W_m H_m C_m} \sum_{i=1}^{W_m} \sum_{j=1}^{H_m} \sum_{k=1}^{C_m} \left(BN_{i,j,k}^{1,m} - BN_{i,j,k}^{2,m} \right)^2 \right] \quad (3)$$

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where $BN^{p,m}$ is the output of the batch normalization that is applied on the layers *Conv* m-4 ($m = 3, 4, 5$) of VGG subnetworks with I_p ($p = 1, 2$) as the input shown in Fig. 1. The subscripts (i, j, k) represent the element position of $BN^{p,m}$. We use W_m, H_m, and C_m to denote width, height, and number of channels, respectively, of the batch normalization layers $BN^{1,m}$ and $BN^{2,m}$. The influence of semantic content features on the loss function L_{S^3N} is controlled by the constants $\alpha_m (m = 3, 4, 5)$, respectively, in the training process.

The new loss function L_{S^3N} measures the difference of semantic content that is the deep-level features between the input I_1 and I_2. In (3), $BN^{1,m}$ and $BN^{2,m}$, $m = 3, 4, 5$, are from *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 in each scale. As a result, the weights of the layers for deep-level features, i.e., *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, are trained to optimize L_{S^3N} directly. The layers for low-level features in the proposed S^3N are not linked directly with $BN^{1,m}$ and $BN^{2,m}$, $m = 3, 4, 5$ and thus L_{S^3N}. The weights of them in the proposed S^3N can be frozen for the low-level features in the minimization of L_{S^3N}. Therefore, L_{S^3N} separates the low- and deep-level features for transfer learning.

In the transfer learning, only the weights of layers for deep-level features are trained to optimize L_{S^3N} based on the separation of the low- and deep-level features. The weights of layers for low-level features are frozen from the pretrained VGG subnetworks. As can be seen in Fig. 1, among the sixteen convolutional layers in the pretrained VGG subnetwork, only three layers, i.e., *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, are adjusted to extract the deep-level features. As the fully connected layers are not adopted in the VGG subnetworks shown in Table I, the total number of parameters of the proposed S^3N is approximately $2 \times 18.76\text{M} = 37.52\text{M}$. The number of parameters to be trained, i.e., the weights of the layers *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 of the two VGG subnetworks, is $2 \times (600\text{K} + 2.3\text{M} + 2.3\text{M}) = 10.4\text{M}$ [38]. The parameters trained in the proposed transfer learning strategy account for merely $(10.4\text{M}/37.52\text{M}) \approx 27.7\%$ of total parameters of the proposed S^3N. Therefore, in the online training, more than 72% of parameters of the proposed S^3N do not need to be trained, leading to corresponding 72% reduction of computational cost for the backpropagation training procedure. As a result, the computational efficiency is greatly improved by the proposed transfer learning strategy.

As the training parameters of the proposed S^3N are limited to the weights of layers for deep-level features, i.e., *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 of two VGG subnetworks, the demand for training samples is reduced dramatically. As shown in Fig. 2, the pairs of unchanged patches in I^{pre} and I^{post} represent the same semantic content of heterogeneous remote sensing images. If these pairs of unchanged patches are input as I_1 and I_2 for semisupervised training, the target output of L_{S^3N} should approach 0 because they share the same semantic content. Therefore, by minimizing L_{S^3N} on these pairs of patches with only unchanged labels, the proposed S^3N suppresses the output difference of $BN^{1,m}$ and $BN^{2,m}$, $m = 3, 4, 5$, in the unchanged regions and thus highlights the regions of change between I^{pre} and I^{post}.

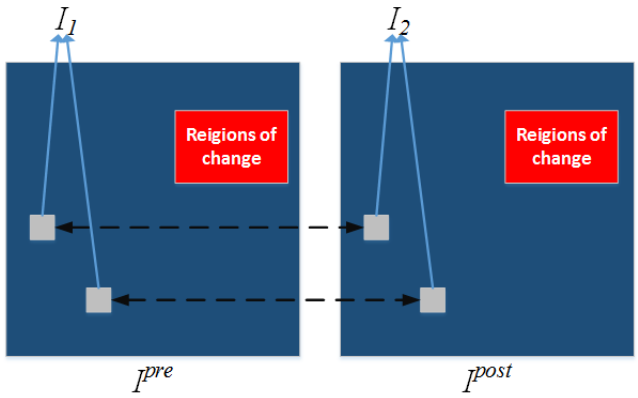


Fig. 2. Selection of training samples of the proposed S^3N for semisupervised training. The red and blue areas indicate the regions of change and nonchange based on ground truths, respectively. The two pairs of patches covered with the gray color in the regions of nonchange, each of which corresponds to the same positions of I^{pre} and I^{post}, are collected as the training samples, i.e., the input I_1 and I_2 of Fig. 1. As mentioned in [21], these samples are very weak prior knowledge.

Compared with the methods [22]–[25] that perform the time-consuming training from scratch on whole images, the proposed S^3N limits the training parameters to the network weights of deep-level semantic content, i.e., *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, leading to great reduction of computational cost. Similar to the homogeneous pixel transformation (HPT) method [21], we utilize the image patches selected from the unchanged regions for training as the unchanged samples. Compared with the HPT method, the proposed S^3N separates the low- and deep-level features. Only the weights of the layers for deep-level features are trained in the proposed transfer learning strategy. It leads to a significant reduction of the number of training parameters. As the number of training parameters decreases, the proposed S^3N needs fewer unchanged samples for training than the HPT method. As a result, the proposed S^3N achieves significant computational efficiency with the novel transfer learning strategy.

The novel transfer learning strategy is summarized in Algorithm 1. The proposed S^3N maps the heterogeneous remote sensing images I^{pre} and I^{post} into the homogeneous feature space based on the deep-level features of semantic content and the low-level features that are transferred accurately from the pretrained VGG network. As a result, direct comparison by difference operation is facilitated in the following inference of change detection map.

D. Inference of Change Detection Map

In this section, we first derive the difference map I_d from I^{pre} and I^{post}. Then, I_d is thresholded to generate the final binary map of change detection I_b.

After the transfer learning, the proposed S^3N achieves appropriate network parameters for change detection in the heterogeneous remote sensing images I^{pre} and I^{post}. Therefore, the inputs of the proposed S^3N in Fig. 1 can be replaced

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Algorithm 1 Training Procedure of the Proposed S^3N

Input:

Preevent image: I^{pre}

Post-event image: I^{post}

Output:

Well-trained S^3N that manages semantic content features of I^{pre} and I^{post} for homogeneous transformation

1: Constructing the proposed S^3N based on the pretrained VGG network [38] on ImageNet according to Fig. 1.
2: Building the loss function according to (3).
3: Choosing a few pairs of patches in the unchanged regions as the inputs I_1 and I_2 with data extension, as shown in Fig. 2.
4: Using the *Adam* optimizer [45] to minimize the loss function in (3) to derive the well-trained S^3N. In the optimization, only the weights of *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 of two VGG subnetworks are adjusted while others are directly transferred from the pretrained VGG networks.

by I^{pre} and I^{post} to infer the difference map

$$I_1 = I^{pre}, \tag{4}$$

$$I_2 = I^{post}. \tag{5}$$

For each scale of the semantic content features, the difference between I^{pre} and I^{post} is given by the direct comparison of the BN layers

$$D_m(i,j) = \frac{1}{C_m}\sum_{k=1}^{C_m}\left(BN_{i,j,k}^{1,m} - BN_{i,j,k}^{2,m}\right)^2 \tag{6}$$

where the subscript $m \in \{3,4,5\}$ indicates the scale levels that share the same meanings with (3), (i,j) represents the coordinates of width and height of D_m, and k refers to the channel of the BN layers. Next, the difference map I_d is generated as follows:

$$I_d = \alpha_3 D_3 + \alpha_4 G(D_4) + \alpha_5 G(D_5) \tag{7}$$

where $G(\cdot)$ is the function that upscales D_4 and D_5 to be the same size with D_3 by bilinear interpolation, and $\alpha_m (m = 3,4,5)$ are the same as (3) in the inference process.

Based on the training procedure on the pairs of unchanged patches selected from I^{pre} and I^{post}, the proposed S^3N maps I^{pre} and I^{post} into the homogeneous feature space of semantic content: $t_1(I^{pre}) = \{BN^{1,m}, m = 3,4,5\}$ and $t_2(I^{post}) = \{BN^{2,m}, m = 3,4,5\}$, respectively, as shown in Fig. 1. For the unchanged regions, the difference between $BN^{1,m}$ and $BN^{2,m}$ is close to 0 after the transfer learning. In the changed regions, the difference is much larger because of their different semantic content. In (6), the pixel-wise difference between $t_1(I^{pre})$ and $t_2(I^{post})$ measured at the third, fourth, and fifth scales of the proposed S^3N is then combined in (7) to derive the difference map, I_d.

Finally, *Otsu* is applied on I_d to derive the binary map of change detection I_b as follows:

$$I_b = f_b(I_d) = Otsu(I_d) \tag{8}$$

Algorithm 2 Inference Procedure of Change Detection Map

Input:

Preevent image: I^{pre}

Post-event image: I^{post}

Well-trained network for homogeneous transformation: S^3N

Output:

Binary map of change detection: I_b

1: Inputting I^{pre} and I^{post} into the well-trained S^3N as I_1 and I_2, according to (4) and (5).
2: Producing the difference of semantic content features in multiple scales of the proposed S^3N, according to (6).
3: Combining the feature difference across the scales derived in step 2 to generate the difference map I_d, according to (7).
4: Employing *Otsu* on I_d to achieve the final binary map of change detection I_b, according to (8).

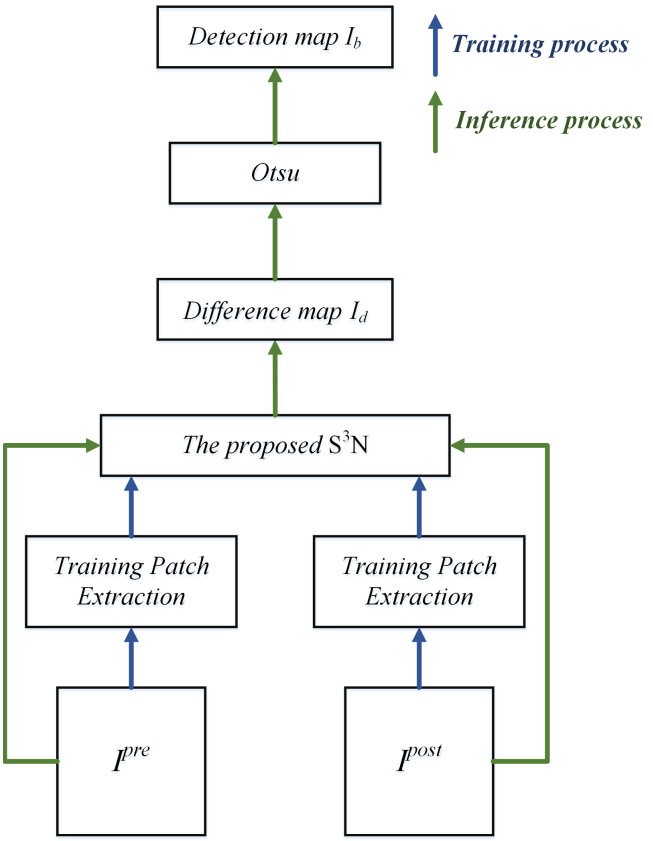


Fig. 3. Flowchart of the proposed S^3N. The arrowed lines with different colors show different procedures of the proposed S^3N: blue arrowed lines indicate the training process; green arrowed lines imply the inference step.

where $Otsu(\cdot)$ denotes the thresholding operation with connected component analysis. The reason for utilizing such a simple thresholding method is to verify that the performance improvements of change detection stem from the proposed S^3N with the novel transfer learning strategy and to exclude the influence of other factors. The inference of binary map of change detection is concluded in Algorithm 2.

In summary, we show the flowchart of the proposed S^3N in Fig. 3.

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E. Analysis of Computational Cost

For the methods based on online training, the computational complexity can be roughly estimated at $O(N_p \times N_i)$, where N_p is the number of training pixels and N_i is the number of iterative steps for learning.

The existing methods based on homogeneous transformation suffer from high computational cost. In [20] and [21], N_p is the number of unchanged pixels chosen for training in the heterogeneous images. To map the pixel values, the essential sorting operations in K-nearest neighbors techniques render the number of iterative steps N_i to be at least $N_p \log N_p$ and thus the computational complexity is $O(N_p^2 \log N_p)$. The training samples selected for mapping the low-level pixel values depend on image sizes. As a result, the expansion of remote sensing image sizes increases the computational cost greatly.

In [23]–[25], the networks with Siamese structures reduce the number of iterative steps N_i due to the symmetric structure in the heterogeneous remote sensing images with the same/symmetric channels. However, the number of training samples N_p is quite large because these methods employ entire images to train the Siamese networks. Therefore, the computational cost for online training is considerably high.

Compared with the existing methods based on homogeneous transformation, the proposed S^{3}N reduces both the number of training samples N_p and iterative steps N_i, leading to a significant decrease in computational cost. By the separation of low- and deep-level features, the proposed S^{3}N avoids the training of the weights of the layers for low-level features. The decrease in the number of network parameters to be trained leads to the reduction of the number of training pixels N_p. Moreover, the number of iterative steps N_i required for online training can be reduced by the Siamese network of the proposed S^{3}N with a symmetric structure. Therefore, the computational cost with the complexity $O(N_p \times N_i)$ is reduced. Furthermore, as the number of network parameters to be trained is limited to the constant number of weights of the layers for deep-level features in the proposed S^{3}N, the demand for training samples and iterative steps is stable and does not depend on the sizes of input images, leading the computational cost robust to the size variation of input images.

III. Experimental Results

In this section, three data sets corresponding to the change events of natural hazards are utilized to evaluate the performance of the proposed S^{3}N. Each of the three data sets consists of a pair of heterogeneous remote sensing images: one collected before the change and the other acquired after. The data sets contain multiple types of remote sensing images with different sizes, including panchromatic, multispectral, SAR, PolSAR, and NDVI images, to validate the proposed S^{3}N. We employ a visual selection of controlling points [46] to coregister the heterogeneous remote sensing images and equalize the resolutions of them by bilinear interpolation, as mentioned in Section II-A. The three data sets are detailed as follows.

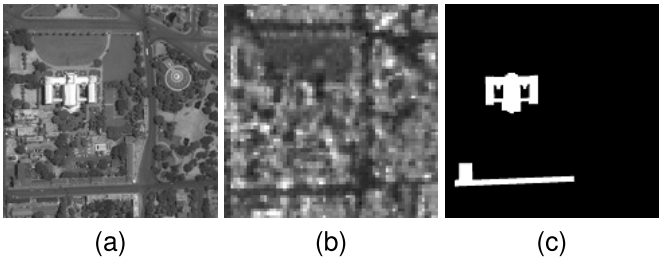


Fig. 4. Heterogeneous remote sensing images of the first data set. (a) Preevent panchromatic image. (b) Post-event SAR image. (c) Ground truth.

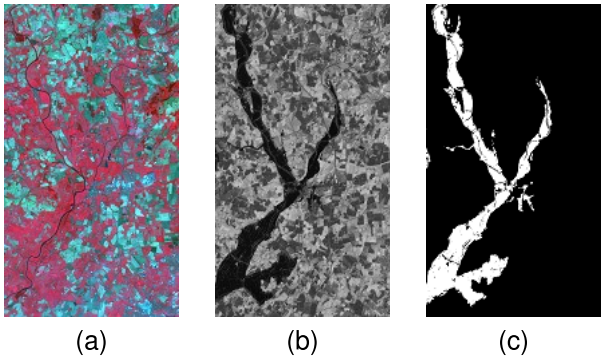


Fig. 5. Heterogeneous remote sensing images of the second data set. (a) Preevent multispectral image. (b) Post-event NDVI image. (c) Ground truth.

In Fig. 4, the first data set comes from a metropolitan area of Port-au-Prince, the capital of Haiti, destroyed severely in the 2010 Haiti earthquake (happened on January 12, 2010). The Quickbird satellite produces the preevent panchromatic image in 2009 and the COSMO-SkyMed satellite provides the post-event SAR image right after the earthquake. United Nations Institute for Training and Research gives the ground truth shown in Fig. 4(c) [48]. The size of heterogeneous images is 640×640 pixels.

The second data set with the ground truth is from [21] to monitor the flooded areas in Gloucester, U.K. on October 21, 1999, as shown in Fig. 5. In Fig. 5(c), vast regions along the river are flooded. The pre-event multispectral image is acquired by the Satellites Pour l'Observation de la Terre (SPOT) satellite in 1999 and the post-event image is the NDVI image with single channel collected in 2000. The size of the two images is 2359×1318 pixels, much larger than that of the first data set. Similar to [21], three spectral bands of SPOT images are considered to facilitate image processing.

The third data set captures the landslide [47] occurred in Xinmo Village, Mao County, Sichuan Province of China on June 24, 2017, as shown in Fig. 6. Before the landslide, a pre-event image shown in Fig. 6(a) is collected by Gaofen-3 satellite in a fully polarimetric model (PolSAR). The post-event multispectral image (similar to the first data set, three spectral bands are considered) is obtained by Gaofen-2 satellite, as shown in Fig. 6(b). The size of the two images is 2827×1333 pixels, slightly larger than that of the second data set. Both of the two images are with the same three channels. National Disaster Reduction Center of China (NDRCC) provides the binary image of ground truth shown in Fig. 6(c).

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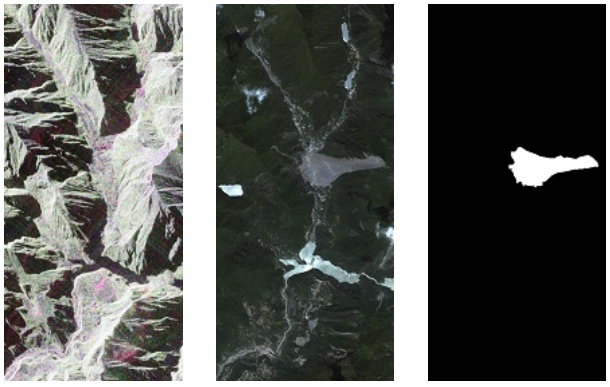


Fig. 6. Heterogeneous remote sensing images of the third data set. (a) Preevent PolSAR image. (b) Post-event multispectral image. (c) Ground truth.

In the experiments, the proposed S^{3}N is compared with three SOTA methods: HPT [21], SCCN [23], and deep homogeneous feature fusion (DHFF) [25], all of which are specialized in change detection in heterogeneous remote sensing images. We also test the method that optimizes all the network parameters of S^{3}N, which is named All-parameter-optimized Siamese Network (ASN), for further validating the effectiveness of the proposed S^{3}N with the transfer learning strategy. HPT maps pixel values between heterogeneous remote sensing images by a multivalue estimation trained on selected pairs of unchanged patches. SCCN uses stacked Denoising AutoEncoder (sDAE) to initialize a Siamese network for homogeneous transformation and derives the final binary map of change detection by minimizing a regularized loss function. DHFF adjusts network weights iteratively to update the transformed homogeneous image for change detection. SCCN and DHFF are the SOTA unsupervised methods that both achieve remarkable performance with Siamese networks. As the proposed S^{3}N also incorporates the Siamese network, the comparison with the two methods illustrates the innovation of change detection in heterogeneous remote sensing images with Siamese structures. In the proposed S^{3}N, the separation of low- and deep-level features of the Siamese network facilitates the following transfer learning strategy and thus achieves significant improvements of detection efficiency based on a small amount of prior knowledge (two image patches, each of which contains no more than 0.25% of unchanged pixels of the entire image). Moreover, the HPT method, which is the weakly supervised SOTA method, is compared with the proposed S^{3}N in the experiments, demonstrating the effectiveness and efficiency of the proposed S^{3}N. The four methods, HPT, SCCN, DHFF, and ASN, need to learn both deep- and low-level features, resulting in inefficient online training procedures with high computational cost.

Both the efficiency and detection performance are evaluated quantitatively. As there exist the unsupervised methods (SCCN and DHFF), of which the training and inference procedures are indivisible, we compare the total running time of the above five methods averaged by over 20 trials. It is fair to compare the total running time because the inference procedures of HPT and the proposed S^{3}N are simple and time saving (by Fuzzy C-means and *Otsu*, respectively). Four measurements regularly used for evaluation of change detection performance are employed, including the accuracy R_a, the precision R_p, the recall R_r, and the Kappa index Ka that are calculated by

$$R_a = \frac{n_a + n_c}{N}, R_p = \frac{n_a}{N_d}, R_r = \frac{n_a}{N_c}, Ka = \frac{R_a - p_e}{1 - p_e} \quad (9)$$

where n_a and n_c are the numbers of changed and unchanged pixels, respectively, with correct detection; N, N_d, and N_c are the total number of pixels of entire image, the number of pixels detected as change, and the number of truly changed pixels, respectively; p_e is the hypothetical probability of random agreements [49]. Besides, as HPT and the proposed S^{3}N are based on semisupervised training, we show the training samples selected by them for comparison. The comparative ASN method employs the same training samples as the proposed S^{3}N.

To conduct the experiments, a server equipped with an Intel Core i9-7980XE CPU, DDR4 RAM of 128 GB, and an NVIDIA TITAN RTX graphics card is utilized as the hardware platform. All codes of the experiments are run in Python 3.5.2 and TensorFlow 1.14 on Ubuntu 16.04.

In what follows, we first provide the training details of the proposed S^{3}N. Then, the experimental results derived by the above five methods are compared and discussed.

A. Training Details

1) Network Settings: To minimize the loss function in (3), *Adam* optimizer is employed. For all the three data sets, the learning rate of *Adam* optimizer is set to be 0.001 and the hyperparameters (β_1, β_2) controlling the exponential decay rates are chosen to be $(0.9, 0.999)$, as the most common and default settings. As the proposed S^{3}N reduces the demand for training samples, we can afford the batch gradient descent (BGD) [50] method to cover all the image patches in one batch for training the proposed S^{3}N, which improves the convergence of the loss function L_{S^3N} in (7) and thus the detection performance. We set the number of training epochs to be 12000 that is sufficiently large to guarantee convergence. Besides, to accommodate the kernel size of *Conv* 1-1, the remote sensing images with one channel, i.e., the SAR image in the first data set and the NDVI image in the second data set, are duplicated into three channels, similar operations to those in [51] and [52].

2) Number of Training Samples: To train the proposed S^{3}N, some pairs of patches are selected as the training samples in each data set, as shown in Fig. 2. Data extension is applied for sufficient training. In each pair, the selected two patches are rotated (90, 180, 270) degrees clockwise and flipped vertically, horizontally, and both vertically and horizontally.

To analyze the relationship between the performance of the proposed method and the number of selected unchanged training samples, we choose different numbers of patches to train the proposed S^{3}N. In each data set, we select one, two, and three pairs of unchanged patches for comparison from two heterogeneous remote sensing images as training samples.

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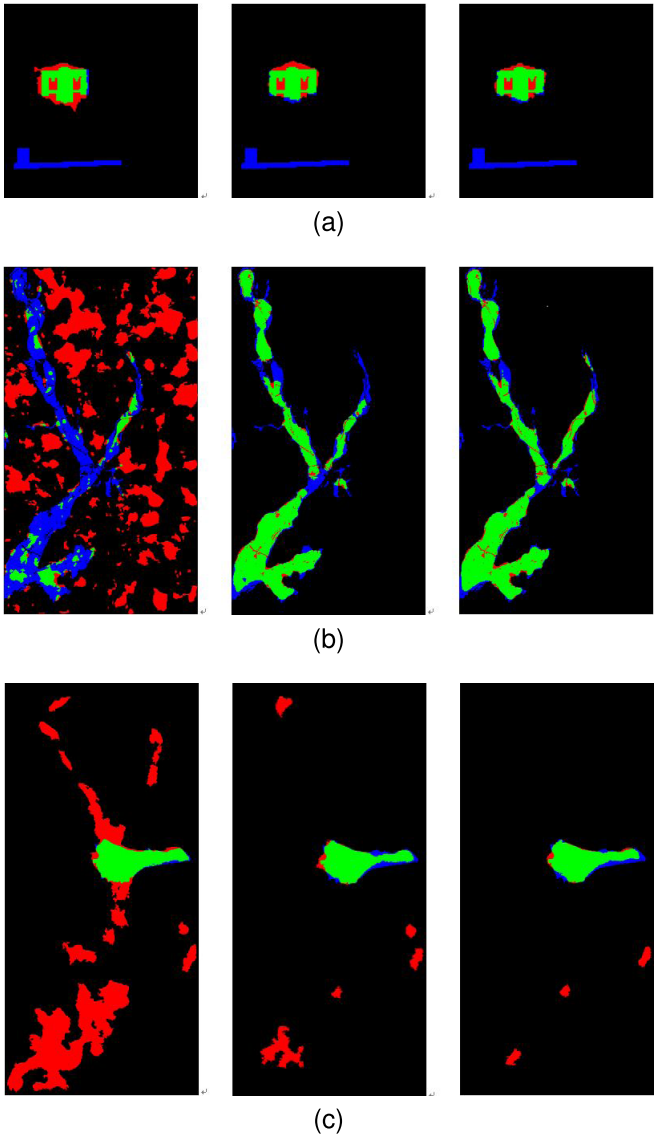


Fig. 7. Detection results (error maps) by the proposed S^{3}N based on different numbers of training samples. (a) First data set. (b) Second data set. (c) Third data set. Different columns indicate different numbers of training samples: the first, second, and third columns are generated based on one, two, and three pairs of training patches, respectively. From left to right, the detection performance is improved with the growing number of training samples.

In each patch, no more than 0.25% of pixels are selected from the entire image, which is merely a small amount of prior knowledge. In the first data set, the size of each patch is 32×32. As the sizes of remote sensing images in the second and third data sets grow much larger, the size of each training patch is increased to be 100×100. We set $(\alpha_3, \alpha_4, \alpha_5) = ((1/3), (1/3), (1/3))$ commonly for a fair comparison while the discussion of parameters settings is provided later in Section III-A3. The detection results, including error maps and bar charts, are provided in Figs. 7 and 8, respectively.

As can be seen in Figs. 7 and 8, the detection performance is improved with growing numbers of training samples. The reason is that the deep-level features for homogeneous transformation are learned more accurately from increasing numbers of training samples. For the results of the third data set shown in Fig. 8(c), although the recall rate R_r (yellow bar) decreases slightly with growing numbers of training samples, the precision rate R_p (green bar) increases rapidly and the overall accuracy R_a (blue bar) that measures the detection performance chiefly is also improved. Therefore, the detection performance is still improved with the growth of training samples in the third data set. For all the three data sets, although the overall detection performance improves with the increasing number of training samples, the computational cost of online training increases, leading to longer running time, as shown in the right column of Fig. 8.

Compared with the performance improvements in the first data set shown in Fig. 7(a), the improvements in the second and third data sets shown in Fig. 7(b) and (c), respectively, are more significant. It is because the sizes of images and training patches in the last two data sets are much larger than those in the first data set. As a result, the increasing number of training patches gives more training samples/pixels to homogeneous transformation for learning the deep-level features, leading to better detection performance.

Based on the above analysis, we choose two pairs of patches for training the proposed S^{3}N, i.e., the middle column of Fig. 7. Compared with other choices, it achieves satisfactory detection performance with reasonable time cost.

3) Settings of Parameters $(\alpha_3, \alpha_4, \alpha_5)$: In the proposed S^{3}N, α_3, α_4, and α_5 control the influence of different semantic content features derived from *Conv* 3-4, *Conv* 4-4, *Conv* 5-4, respectively, on the detection performance, as shown in (3) and (7). Without loss of generality, we assume that $\alpha_3 + \alpha_4 + \alpha_5 = 1$ and select different $(\alpha_3, \alpha_4, \alpha_5)$ to evaluate the overall detection performance: R_a and Ka. Table II shows the values of R_a and Ka versus different $(\alpha_3, \alpha_4, \alpha_5)$. In Table II, the best detection performance is achieved when α_3, α_4, and α_5 are unequal in each data set, which means that the contributions of each semantic content features to the detection performance are different. As shown in Table II, lower α_5 gives better overall detection results, indicating that the influence of α_5 is less than that of α_3 and α_4. It is probably because the resolution of *Conv* 5-4 is decreased considerbly by pooling operations. Different from the first two data sets, the third data set gives the best performance when α_3 is set to be lower ($\alpha_3 = 0.3$). The main reason is that the radar image distortions that cause massive false alarms. As can be seen in Fig. 15(a), the detection noise in D_3 derived by *Conv* 3-4 is more serious than that in the first two data sets, shown in Fig. 9(a) and Fig. 12(a). Therefore, the importance of semantic features derived by *Conv* 3-4 is decreased in the third data set. Specifically, we set $(\alpha_3, \alpha_4, \alpha_5) = (0.4, 0.4, 0.2)$ as it achieves satisfactory detection performance on all of the three data sets, including the best average R_a and the second best average Ka.

B. Experimental Results on the First Data Set

In the first data set, Fig. 9 presents the visualization results derived by the proposed S^{3}N. Fig. 10 provides the error maps of change detection and Table III gives the corresponding quantitative evaluations. The samples selected for training HPT and the proposed S^{3}N are both depicted in Fig. 11. The

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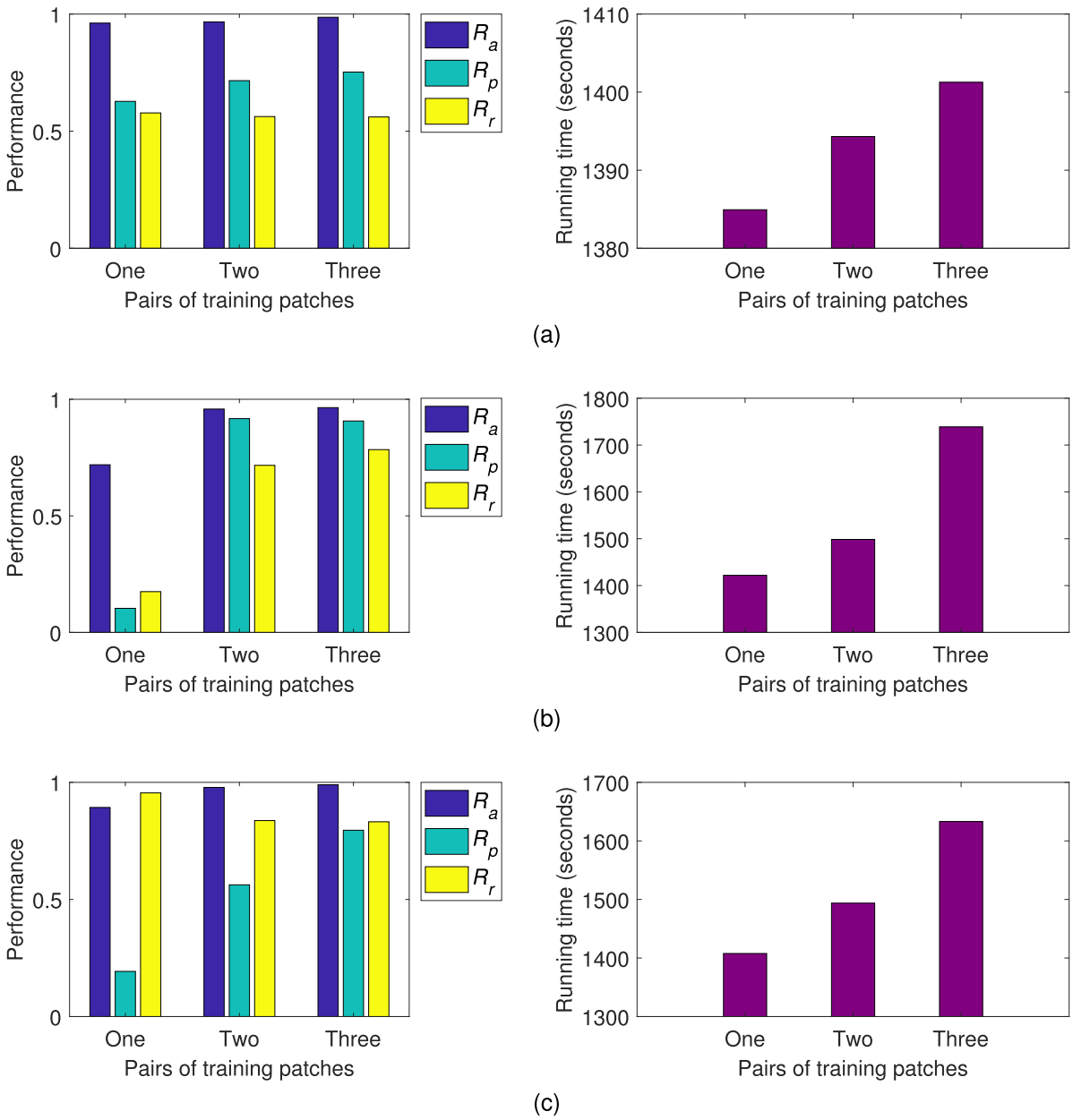


Fig. 8. Comparisons of detection performance (left column) and running time (right column) across different numbers of training samples. (a) First data set. (b) Second data set. (c) Third data set. In the left column, R_a (blue bar), R_p (green bar) and R_r (yellow bar) represent the overall accuracy, precision rate, and recall rate, respectively. For all the three data sets, with the growing number of training samples, the proposed S^{3}N achieves overall better performance but needs more running time.

image size of the first data set is the smallest among the three experimental data sets.

In Fig. 9(a), the details of changed regions are clear. It is because the semantic features derived by *Conv* 3-4 focus more on the descriptions of image details. As the semantic features derived by *Conv* 4-4 and *Conv* 5-4 go deeper, the integrity of detected regions is improved while the edges of changed regions become obscure, shown in Fig. 9(b) and 9(c). It illustrates that the deeper semantic features provide the detection results with more integrity. By fusing the difference of semantic features derived by *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4 in (7), the difference map I_d shown in Fig. 9(d) preserves both integrity and details of the detected regions, demonstrating the effectiveness of the proposed S^{3}N with the novel transfer learning strategy.

As can be seen in Fig. 10 and Table III, there exist massive false alarms in the detection results of HPT, SCCN, DHFF, and ASN. It is because these methods do not learn the homogeneous transformation well in these regions with the same semantic content between heterogeneous remote sensing images. The proposed S^{3}N achieves a better precision rate R_p and fewer false alarms than the other comparative methods: HPT, SCCN, DHFF, and ASN, illustrating the effectiveness of the proposed S^{3}N with the novel transfer learning strategy. As can be seen in Fig. 10(d), the detection performance of the ASN method is unsatisfactory with massive false alarms and

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TABLE II
DETECTION PERFORMANCE OF THE PROPOSED S^{3}N WITH DIFFERENT $(\alpha_3, \alpha_4, \alpha_5)$

$(\alpha_3, \alpha_4, \alpha_5)$	R_a/Ka			
	First dataset	Second dataset	Third dataset	Average
$(0.2, 0.3, 0.5)$	$0.9622/0.5765$	$0.9450/0.6934$	$0.9816/0.6901$	$0.9629/0.6533$
$(0.3, 0.2, 0.5)$	$0.9631/0.5758$	$0.9379/0.6419$	$0.9791/0.6606$	$0.9600/0.6261$
$(0.2, 0.5, 0.3)$	$0.9646/0.6032$	$0.9624/0.8087$	$0.9813/0.6898$	$0.9694/0.7006$
$(0.3, 0.5, 0.2)$	$0.9664/0.6121$	$0.9645/0.8233$	**0.9823**/**0.6986**	0.9711/**0.7113**
$(0.5, 0.2, 0.3)$	$0.9694/0.6168$	$0.9587/0.7879$	$0.9799/0.6750$	$0.9693/0.6932$
$(0.5, 0.3, 0.2)$	**0.9697**$/0.6218$	$0.9629/0.8160$	$0.9792/0.6627$	$0.9706/0.7002$
$(0.2, 0.4, 0.4)$	$0.9629/0.5921$	$0.9564/0.7691$	$0.9801/0.6792$	$0.9665/0.6801$
$(0.4, 0.2, 0.4)$	$0.9663/0.5978$	$0.9526/0.7436$	$0.9581/0.5044$	$0.9590/0.6153$
$(0.4, 0.4, 0.2)$	0.9688/**0.6226**	**0.9651**/**0.8281**	$0.9796/0.6712$	**0.9712**$/0.7073$
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	$0.9665/0.6114$	$0.9581/0.7815$	$0.9783/0.6620$	$0.9676/0.6850$

The **boldface** indicates the best results.

TABLE III
COMPARISONS OF DETECTION METHODS ON THE FIRST DATA SET

Method	R_a	R_p	R_r	Ka	Running time
HPT	0.9365	0.3722	0.3883	0.3467	**622.505s**
SCCN	0.9464	0.4584	0.3867	0.3916	$1718.980s$
DHFF	**0.9823**	0.5819	**0.7623**	**0.6404**	$2759.203s$
ASN	0.8293	0.0971	0.2874	0.0755	$1778.965s$
S^{3}N	0.9688	**0.7672**	0.5467	0.6226	$1376.485s$

The **boldface** indicates the best results.

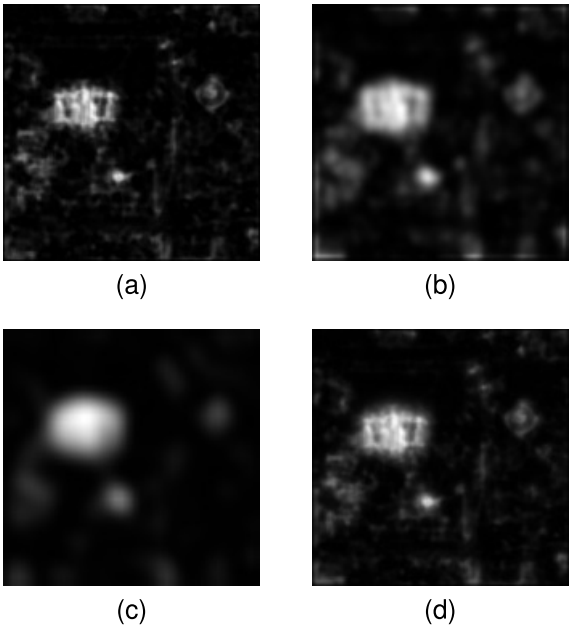


Fig. 9. Visualization of difference of semantic features and difference map of the first data set. (a) D_3 derived by *Conv* 3-4. (b) D_4 derived by *Conv* 4-4. (c) D_5 derived by *Conv* 5-4. (d) Difference map I_d. In (7), the difference of semantic features, i.e., D_3, D_4, and D_5, is combined to derive the difference map, I_d. As can be seen, the difference map I_d preserves both integrity and details of the detected regions of change.

missed targets. It is because the semantic content is corrupted seriously in the homogeneous transformation by optimizing all the network parameters. While the training of the proposed S^{3}N may be a little overfitting, the damaged building in the lower part of the images, shown in Fig. 10(e), misses the detection. The proposed S^{3}N transforms them wrongly into the same homogeneous features. Despite the missed target, the performance of the proposed S^{3}N is still better than that of HPT, SCCN, and ASN in terms of R_a, R_r, and Ka, and only second to that of DHFF.

Table III also provides the running time of comparative methods. Compared with the ASN method that optimizes all the network parameters, the proposed S^{3}N with the transfer learning strategy only trains the deep-level features, leading to a significant reduction of computational cost. Although the DHFF method performs a little better than the proposed S^{3}N on R_a and Ka, it costs more than twice the running time because of the time-consuming update of all the network parameters. Compared with the DHFF method, the proposed S^{3}N is much more time-efficient. In Table III, the time efficiency of the proposed S^{3}N is second to that of HPT due to the small sizes of input images. It is because the requirement of patches for training HPT depends on the sizes of input images, leading to the decrease of computational cost $O({N_p}^2 \log N_p)$. However, as can be seen in Table III, the detection results of HPT are unsatisfied with the lowest R_a, R_r, and Ka. Compared with HPT, the proposed S^{3}N achieves much better detection performance, shown in Fig. 10 and Table III. It validates both efficiency and effectiveness of the proposed S^{3}N based on the novel transfer learning strategy.

The training samples for HPT and the proposed S^{3}N are shown in Fig. 11 with quantitative comparisons. Compared with HPT, the proposed S^{3}N needs approximately one-tenth of training samples. The proposed transfer learning strategy

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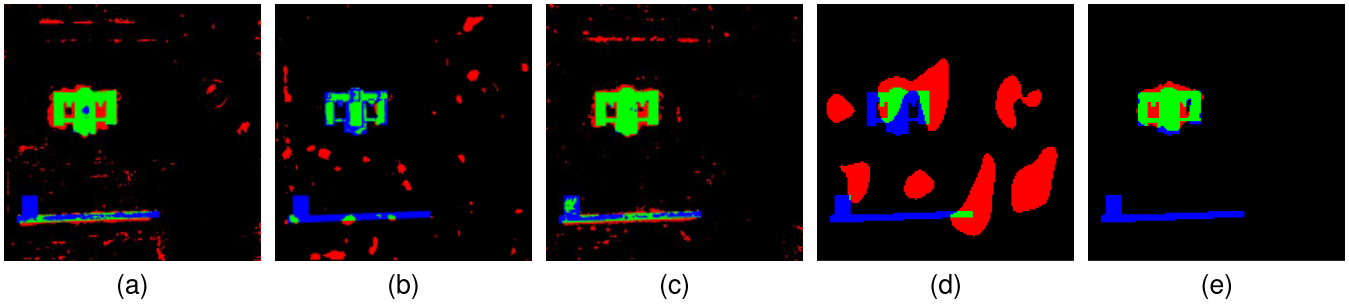


Fig. 10. Comparisons of detection results (error maps) of the first data set, achieved by (a) HPT, (b) SCCN, (c) DHFF, (d) ASN, and the (e) proposed S^{3}N. Here the green, red, blue, and black colors indicate the regions of true positives, false alarms, missed targets, and true negatives. As can be seen, the performance of the proposed S^{3}N (e) matches that of the other methods.

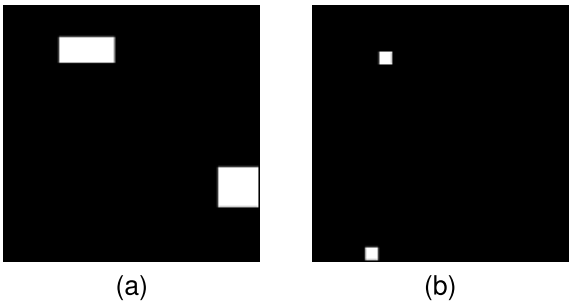


Fig. 11. Sampled regions of the semisupervised methods on the first data set marked with white color. (a) HPT with 19135 training pixels. (b) Proposed S^{3}N with 2048 training pixels. As can be seen, the proposed S^{3}N depends on much fewer training samples.

that only trains the weights of the layers for deep-level semantic content reduces the demand for training samples dramatically.

C. Experimental Results on the Second Data Set

The image size of the second data set is much larger than that of the first data set (The number of pixels is nearly eight times larger than that of the first data set). Fig. 12 visualizes the difference of semantic features derived by the proposed S^{3}N. The binary maps of change detection and the quantitative evaluations, corresponding to the second data set, are presented in Fig. 13 and Table IV, respectively. The selected regions for training the proposed S^{3}N and HPT, both of which are semisupervised, are provided in Fig. 14.

Similar with the first data set, the difference of semantic features becomes abstract as it ranges from *Conv* 3-4 to *Conv* 5-4, as shown in Fig. 12(a), (b), and (c). The main reason is that the abstract ability of the layers in the proposed S^{3}N is improved with the layer depth. The difference map I_d shown in Fig. 12(d) fuses the difference of semantic features derived by *Conv* 3-4, *Conv* 4-4, and *Conv* 5-4, leading to the detection improvements on both integrity and details.

As can be seen in Fig. 13 and Table IV, the proposed S^{3}N performs better than the SOTA methods in the detection performance. In Fig. 13(d), the performance of the ASN method is poor because the optimization of all the network parameters corrupts the semantic content of heterogeneous remote sensing images seriously. This demonstrates the effectiveness of the proposed S^{3}N with the novel transfer learning strategy. Compared with other methods, the proposed S^{3}N transforms the heterogeneous remote sensing images into the homogeneous feature space more accurately. Therefore,

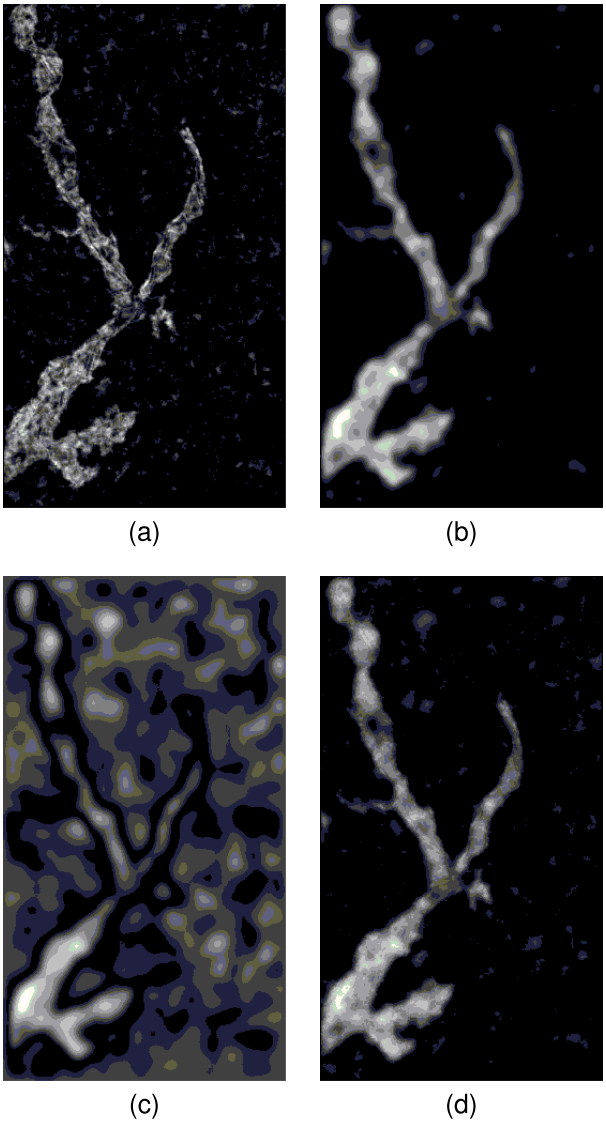


Fig. 12. Visualization of difference of semantic features and difference map of the second data set. (a) D_3 derived by *Conv* 3-4. (b) D_4 derived by *Conv* 4-4. (c) D_5 derived by *Conv* 5-4. (d) Difference map I_d. Similar with Fig. 9, in the difference map I_d, both integrity and details of the detected regions are improved.

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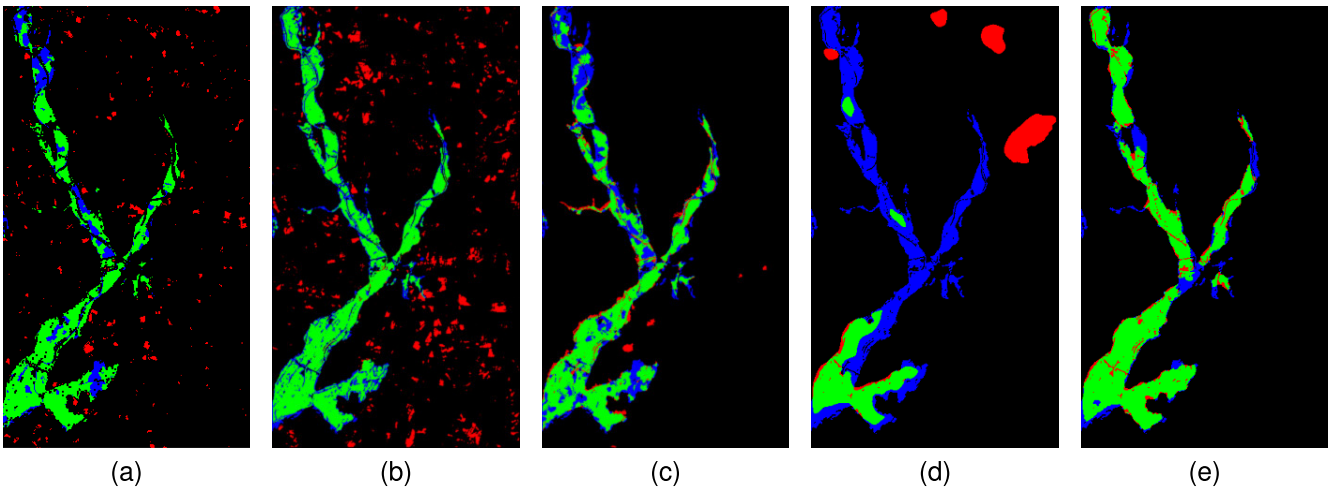


Fig. 13. Comparisons of detection results (error maps) of the second data set, achieved by (a) HPT, (b) SCCN, (c) DHFF, (d) ASN, and the (e) proposed S^{3}N. The image size is much larger than that of the first data set. Similar to the first data set, the green, red, blue, and black colors indicate the regions of true positives, false alarms, missed targets, and true negatives. As can be seen, the performance of the proposed S^{3}N (e) still matches that of the other methods.

TABLE IV
COMPARISONS OF DETECTION METHODS ON THE SECOND DATA SET

Method	R_a	R_p	R_r	Ka	Running time
HPT	0.9641	0.8262	0.7689	0.7761	2590.636s
SCCN	0.9354	0.7064	0.7818	0.7054	19652.450s
DHFF	0.9459	0.8647	0.6469	0.7107	19082.733s
ASN	0.8922	0.6033	0.3002	0.3489	5062.696s
S^{3}N	**0.9651**	**0.8918**	**0.8078**	**0.8281**	**1479.957s**

The **boldface** indicates the best results.

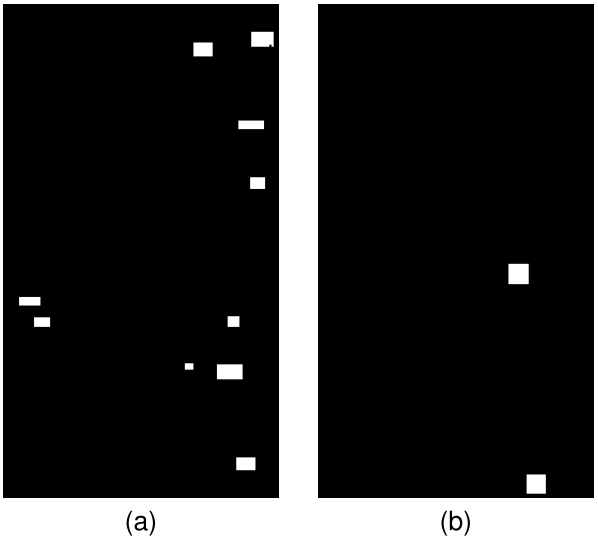


Fig. 14. Sampled regions of the semisupervised methods on the second data set marked with white color. (a) HPT with 54330 training pixels. (b) Proposed S^{3}N with 20000 training pixels. Although the image size of the second data set is much larger than that of the first data set, the proposed S^{3}N relies on much fewer training samples, compared with HPT.

the regions of change and nonchange can be identified by the difference of transformed homogeneous features.

In Table IV, the running time of the proposed S^{3}N is much shorter than that of other methods despite the increase of image sizes. SCCN spends the longest time because the learning of homogeneous features of whole images from scratch prolongs the training procedure greatly. Compared with DHFF, the proposed S^{3}N takes much less running time by the novel transfer learning strategy. The DHFF method randomizes all the network parameters for homogeneous transformation that prolongs the online training procedure greatly. Instead, the novel transfer learning strategy of the proposed S^{3}N adjusts only the weights of the layers for deep-level features in the online training procedure, leading to a significant reduction of the number of training parameters and thus significant improvements of detection efficiency. Compared with SCCN and DHFF, HPT achieves lower computational cost by applying semisupervised learning only on some unchanged regions. However, the computational cost increases rapidly with the image sizes, compared with the first data set. Compared with HPT, the proposed S^{3}N trains the deep-level features for homogeneous transformation to reduce the computational cost without hurting the detection performance. It demonstrates both efficiency and effectiveness of the proposed S^{3}N with the novel transfer learning strategy. Similar to the first data set, as the ASN method optimizes all the network parameters, the computational cost increases greatly. Compared with ASN, the proposed S^{3}N is more time-efficient with less than one-third of the time cost. It illustrates the effectiveness of the proposed transfer learning strategy. Furthermore, the constant parameters to be trained in the proposed S^{3}N prevent the

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increase of computational cost with image sizes. As shown in Tables III and IV, the training time of the proposed S^{3}N grows much slower than that of other methods.

The training samples selected by HPT and the proposed S^{3}N are displayed in Fig. 14. As the novel transfer learning strategy trains only a few network weights of deep-level features, the proposed S^{3}N needs less than half of the training samples of HPT to learn the homogeneous transformation, leading to a significant reduction of computational cost.

D. Experimental Results on the Third Data Set

Different from the size of images in the second data set, the size of images in this data set is slightly larger. Moreover, the pre- and post-event heterogeneous remote sensing images are both colored with three channels. The difference of semantic features derived by the proposed S^{3}N is visualized in Fig. 15, along with the difference map I_d. Fig. 16 and Table V show the change detection results with error maps and quantitative evaluations, respectively. Fig. 17 presents the selected samples for training HPT and the proposed S^{3}N.

In Fig. 15(a)–(c), the difference of semantic features derived by the deeper layers is more abstract, similar to the visualized results shown in Figs. 9 and 12. The proposed S^{3}N assigns different weights to the semantic content features and combines them into the difference map I_d. As a result, both integrity and details of the detected regions are described more accurately, as shown in Fig. 15(d).

In Fig. 16(a), the detection performance of HPT is deteriorated by the increasing image sizes that cause difficulty for learning the mapping of pixel values of entire images. SCCN is with only four layers on each side to extract the homogeneous features that may not be deep enough for the heterogeneous remote sensing images with the increasing sizes. Compared with HPT and SCCN, DHFF is better with the highest R_a and R_p but misses too many targets. The proposed S^{3}N achieves better R_r and Ka than the DHFF method because the proposed S^{3}N only trains the weights of the layers for deep-level features instead of all the network parameters. Compared with the precision rate R_p in the first two data sets, R_p in the third data set is lower with more false alarms. It is because there are more radar image distortions in the pre-event PolSAR image of the third data set. As can be seen in Fig. 6(a) from the third data set, the distortions of massive shadowed regions are on the left side of ridges caused by the side-looking PolSAR imaging mode. The image distortions in Fig. 6(a) increase the difficulty for homogeneous transformation, leading to increased false alarms and thus unsatisfactory R_p. Also note that, for the third data set, the proposed S^{3}N still provides higher R_p and fewer false alarms than the competing methods SCCN and HPT, as shown in Table V. Although the DHFF method achieves better R_p than the proposed S^{3}N, it sets an extremely strict detection threshold to reduce the false alarms, leading to the worst recall rate R_r with massive missed targets among the five methods. Compared with the ASN method that corrupts the semantic content, the proposed S^{3}N achieves better performance because the transfer learning strategy only trains the weights of the layers for deep-level features instead of all the network parameters. Therefore, compared with HPT, SCCN, DHFF, and ASN, the proposed S^{3}N is more robust to the image distortions induced by the PolSAR side-looking mode.

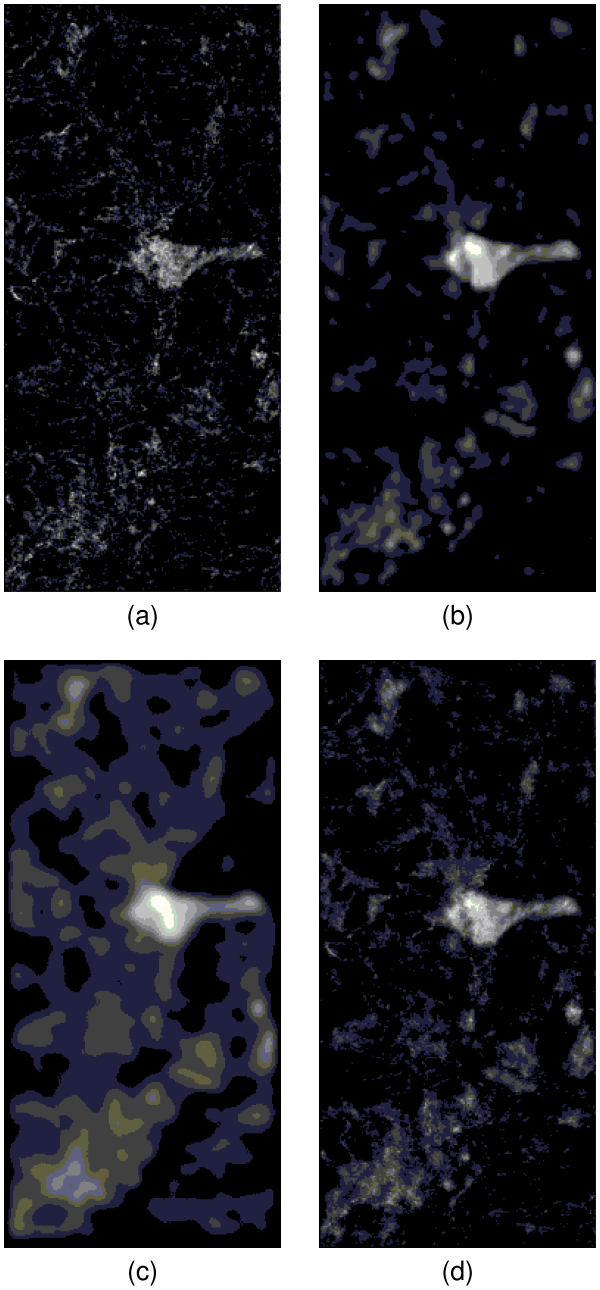


Fig. 15. Visualization of difference of semantic features and difference map of the third data set. (a) D_3 derived by *Conv* 3-4. (b) D_4 derived by *Conv* 4-4. (c) D_5 derived by *Conv* 5-4. (d) Difference map I_d. Similar with Figs. 9 and 12, both integrity and details of the detected regions are improved in the difference map I_d shown in (d) by the fusion in the proposed S^{3}N.

In Table V, both SCCN and DHFF cost the longest running time, compared with other methods. The learning of all the features for homogeneous transformation is the main reason for the efficiency decline. Compared with SCCN and DHFF, the reduction of the running time of HPT is obvious. However, the computational cost is much higher than that of the second data set. It is caused by the increasing sizes of input images.

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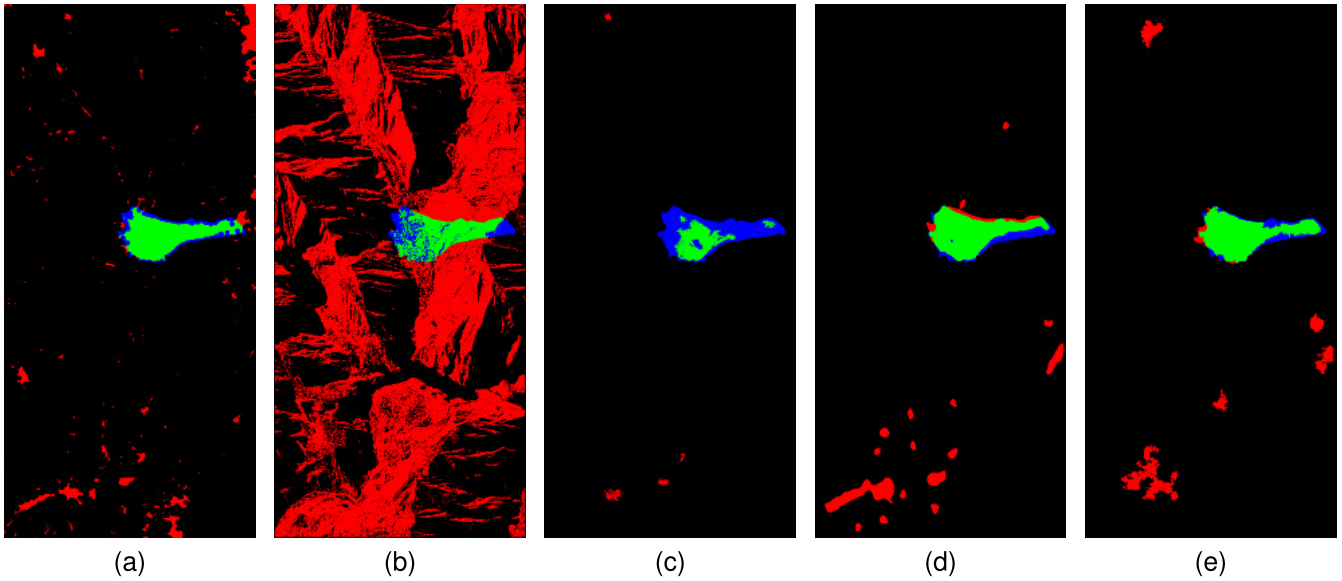


Fig. 16. Comparisons of detection results (error maps) of the third data set, achieved by (a) HPT, (b) SCCN, (c) DHFF, (d) ASN, and the (e) proposed S^{3}N. Same with Figs. 10 and 13, the green, red, blue, and black colors indicate the regions of true positives, false alarms, missed targets, and true negatives. Evidently, the proposed S^{3}N (e) rivals the other methods.

TABLE V
COMPARISONS OF DETECTION METHODS ON THE THIRD DATA SET

Method	R_a	R_p	R_r	Ka	Running time
HPT	0.9627	0.3676	0.5771	0.4307	$3628.848s$
SCCN	0.5709	0.0406	0.6714	0.0280	$13292.981s$
DHFF	**0.9810**	**0.8586**	0.3399	0.4792	$15027.860s$
ASN	0.9779	0.5587	0.8086	0.6498	$4946.391s$
S^{3}N	0.9796	0.5838	**0.8181**	**0.6712**	**1478.565s**

The **boldface** indicates the best results.

Because the ASN method optimizes all the network parameters, its time cost is more than triple that of the proposed S^{3}N. Among the five methods, the proposed S^{3}N is the most computationally efficient. It takes only 41% of the running time of HPT that also runs much faster than SCCN, DHFF, and ASN. The main reason is that the computational cost for learning the low-level features is reduced by the novel transfer learning strategy.

Fig. 17 presents the selected training samples for the semisupervised methods, HPT, and the proposed S^{3}N. The proposed S^{3}N reduces two-thirds of the number of training samples, compared with HPT. It is because the transfer learning strategy limits the network training parameters to only a few weights of the layers for deep-level features for homogeneous transformation, leading to a significant reduction of demand for training samples.

E. Discussions of Computational Cost for Different Data Sets

For different algorithms, the running time is compared across the three data sets with different sizes of heterogeneous remote sensing images, as shown in Fig. 18. As can be seen, the running time of SCCN and DHFF increases extremely fast from the first to the second data set with a great increase of image sizes. The main reason is that both SCCN and DHFF employ entire images to learn all the homogeneous features. The rapid increase of image sizes from the first to the second data set leads to considerably large N_i (number of training samples) and thus high computational cost $O(N_p \times N_i)$. Note that although the image size of the third data set is slightly larger than that of the second data set, the running time of SCCN and DHFF decreases. As mentioned in Section II, it is because the two input images of the third data set are both with three channels, and are more symmetric than the two images of the second data set. The networks reduce the number of iterative steps N_i in the third data set, leading to the overall decrease of computational cost $O(N_p \times N_i)$.

As the ASN method employs the image patches instead of entire images for training, its running time decreases, compared with SCCN and DHFF. However, the optimization of all the network parameters reduces the time efficiency of the ASN method greatly.

Compared with SCCN, DHFF, and ASN, the HPT method is more efficient with less running time. However, the increase of image sizes reduces the efficiency significantly. It is because the number of training samples N_p selected by HPT increases with the sizes of input images, leading to a great increase of iterative steps $N_p \log N_p$ and thus the computational cost $O({N_p}^2 \log N_p)$.

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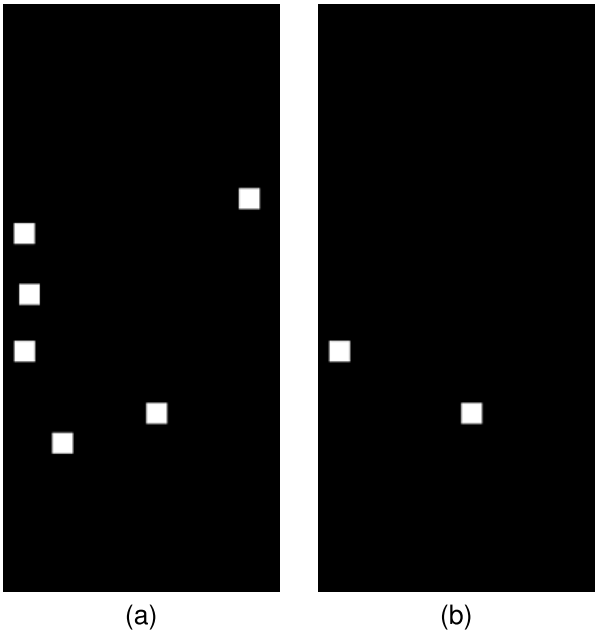


Fig. 17. Sampled regions of the semisupervised methods on the third data set marked with white color. (a) HPT with 60 000 training pixels. (b) Proposed S^{3}N with 20 000 training pixels. Similar to the first and second data set, the proposed S^{3}N relies on much fewer training samples.

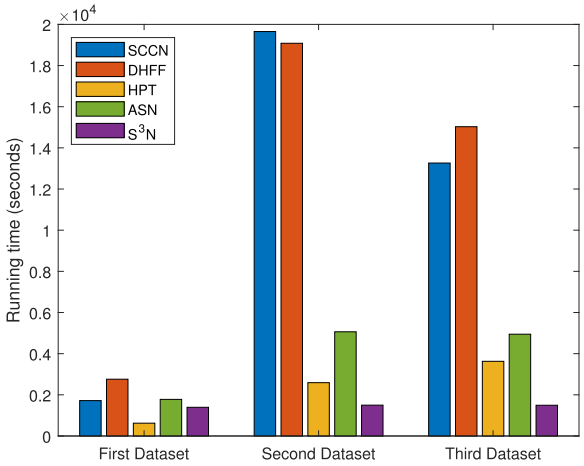


Fig. 18. Comparisons of running time across the three experimental data sets with different image sizes. The x-axis gives the algorithms for comparison on different data sets. The y-axis is the running time, also shown in Tables III–V. As can be seen, the proposed S^{3}N is the most computationally efficient and stable especially with the increase of image sizes.

Compared with all other methods, the proposed S^{3}N is more computationally efficient and also robust to the size variation of heterogeneous remote sensing images. In Fig. 18, the running time of the proposed S^{3}N remains almost constant on the three data sets. Compared with the ASN method that optimizes all the network parameters, the proposed S^{3}N with the transfer learning strategy limits the parameters for training to the weights of the layers for deep-level features, leading to a significant reduction of running time. Note that although the proposed S^{3}N runs a little slower than HPT in the first data set due to the small sizes of input images, it performs much better in the change detection, shown in Fig. 10 and Table III. Furthermore, in the proposed S^{3}N, the Siamese network reduces the number of iterative steps N_i required for online training. Therefore, the computational cost $O(N_p \times N_i)$ is reduced.

IV. Conclusion

In this article, we present a new network, namely S^{3}N, with a novel transfer learning strategy for change detection in heterogeneous remote sensing images. The proposed S^{3}N builds a Siamese structure by employing two VGG networks as the subnetworks. The two VGG subnetworks pretrained well on natural images are further trained on the pairs of patches selected from the heterogeneous remote sensing images. The two VGG subnetworks separate the outputs of the proposed S^{3}N into the low- and deep-level features. In the transfer learning strategy, the network weights of low-level features are frozen from the pretrained VGG networks with no additional training because they hold for remote sensing images identically. Only the deep-level features that are different between natural images and remote sensing images are adjusted in the online training procedure to extract homogeneous features for change detection. As only the deep-level features are learned in the proposed S^{3}N, the novel transfer learning strategy reduces the computational cost significantly.

Compared with the SOTA methods via homogeneous transformation, the proposed S^{3}N is much more computationally efficient with comparable detection performance. The proposed transfer learning strategy eliminates the learning of low-level features in the existing methods, leading to a great reduction of computational cost. By training the deep-level features, the detection performance of the proposed S^{3}N is maintained and still matches that of the SOTA methods. Furthermore, the efficiency of the proposed S^{3}N is robust to the size variation of input images. The constant number of network parameters to be trained to lead the computational cost independent on size variation of image sizes.

The performance of the proposed S^{3}N is evaluated on three real data sets including the heterogeneous remote sensing images of different types and sizes. The experimental results validate both efficiency and effectiveness of the proposed S^{3}N. Despite nearly ten times increase of image sizes ranging from the first to the third data set, less than 8% growth of running time is gained by the proposed S^{3}N, much lower than that of other methods. This illustrates the efficiency of the proposed new S^{3}N with the novel transfer learning strategy. Furthermore, for the Kappa index as the comprehensive assessment of detection performance, the proposed S^{3}N is the highest on the second and the third data sets, and second to the best with only 2% lower in the first data set, proving the effectiveness of the proposed new S^{3}N.

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